

Logic Design and Switching Theory

Lecture – 9

Subtitle: Encoders, Priority Encoders, Magnitude Comparator and Decimal to BCD Encoders.

by
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Previous

Decoder

- 3x8 decoder (with active low and active high outputs)
- 4x16 decoder
- Implementation of logic expression using decoder
- 74138: 3x8 decoder
- Cascading two 2x4 decoder to make 4x16 decoder
- Cascading two 3x8 decoder to make 4x16 decoder
- Cascading two 74138 decoder to make 4x16 decoder
- Cascade 4x16 decoder using 2x4 decoder
- Cascade 5x32 decoder using 4x16 decoder
- Cascade 5x32 decoder using 4x16 decoder
- BCD to 7 Segment code converter
- BCD to decimal Decoder
- Constructing full adder using decoder

MUX

- Implementing logic expression in MUX
- Constructing full adder using MUX

DMUX

Today's Topic

Encoders

- Decimal to BCD encoder
- BCD to Seven segment display encoder

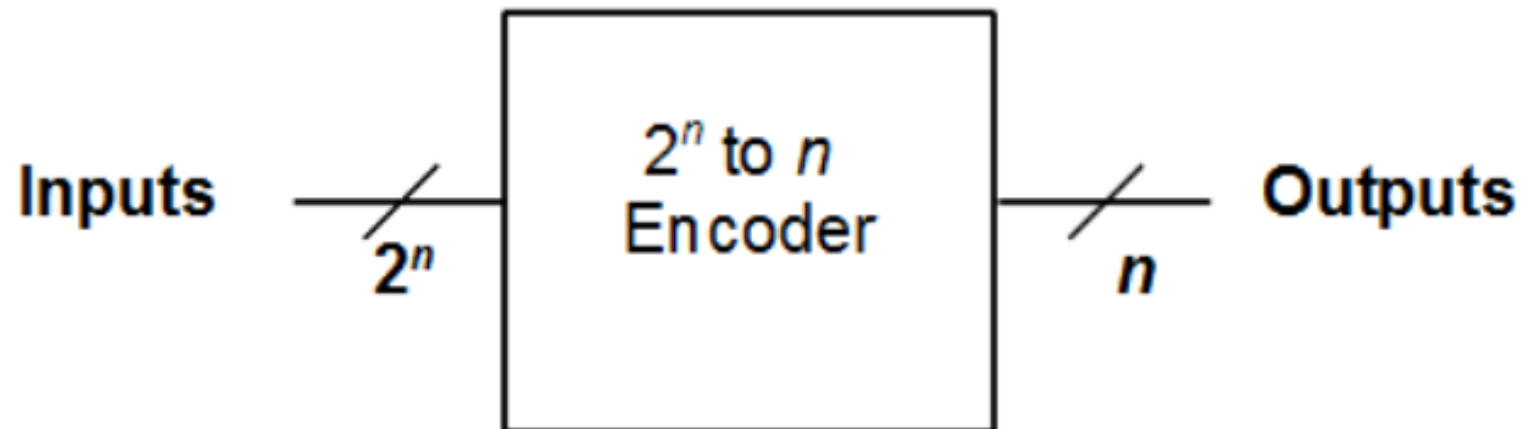
Priority encoders

- Decimal to BCD Priority encoder
- IC:74147
- IC:74148 Octal to Binary Priority encoder
- Cascading two 74148 to form 16x4 priority encoder

Encoders

An encoder is a digital circuit that converts one active input line (out of many) into a binary code representing which input was active. The encoder can have only one input active at a time, otherwise the circuit has no meaning. Simple encoders are not available in ICs as they can simply be converted with or gates.

An encoder is a combinational circuit that converts 2^n input lines into n output lines.



Applications of Encoders

1. Input to Code Conversion

One of the main purposes of using an encoder is to convert input signals into a coded format.

For example, Imagine if you had a calculator keypad. The thing is that when you press a number, say '7', there is a circuit that detects which key was pressed and converts it to a binary value, which would be 0111.

2. Parallel to Serial Conversion

Encoders in Digital Electronics are also useful in converting data from a parallel format to a serial format.

Example: USB keyboards. Within the keyboard, the keypresses are stored in registers in digital electronics following a matrix method (parallel form) however, it is sent serially, over the USB cable, to the computer.

Applications of Encoders

3. Reducing Wiring Complexity

And then when you have to work with multiple inputs and only so many lines to transport that data, things can get very messy in terms of wiring. An encoder can simplify this to some extent.

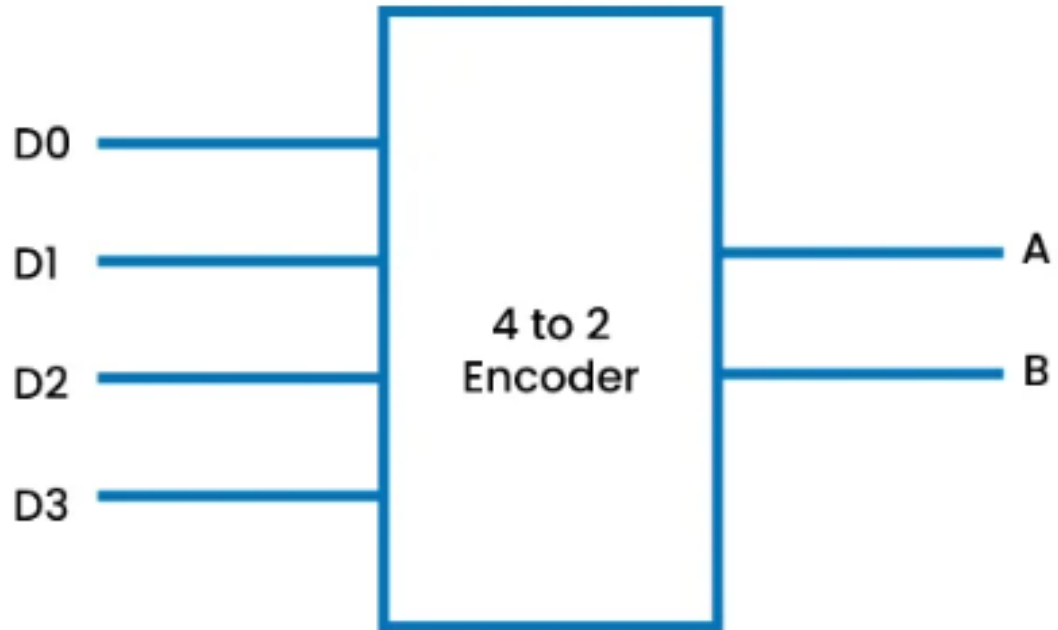
Example: There are multiple buttons or control signals in industrial machines. Rather than running 16 wires to a controller between buttons, we can actually convert which button is pressed to a 4-bit binary value using an encoder. All of these 4 bits are transmitted on only 4 lines. This simplifies the system and makes it easier to debug.

Types of Encoders

There are following types of Encoders in digital electronics.

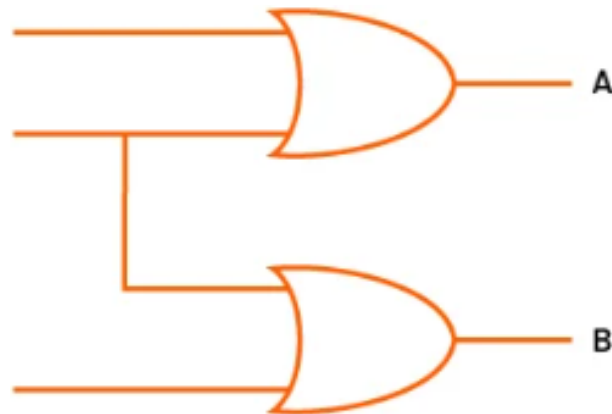
1. 4 to 2 Bit Encoder
2. 8 to 3 Bit Encoder or Octal to Binary Encoder
3. Decimal to BCD Encoder

4x2 line Encoders

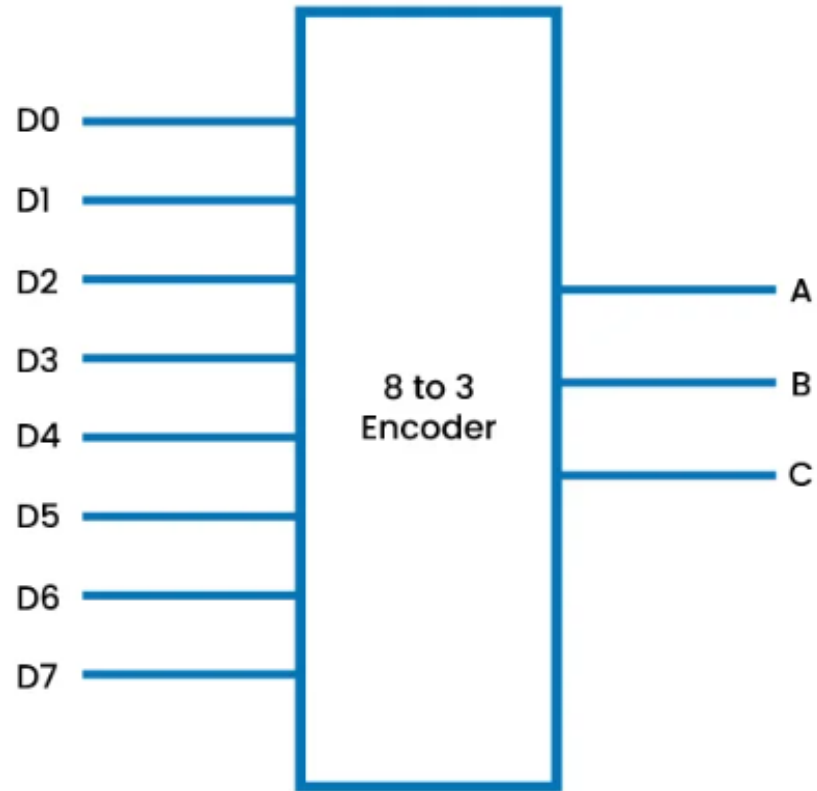


Truth Table

D0	D1	D2	D3	A	B
1	0	0	0		
0	1	0	0		
0	0	1	0		
0	0	0	1		



8x3 Encoders



Truth Table

D0	D1	D2	D3	D5	D6	D6	D7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

8x3 Encoders

D0	D1	D2	D3	D5	D6	D6	D7	A	B	C
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$$A = D4 + D5 + D6 + D7$$

$$B = D2 + D3 + D6 + D7$$

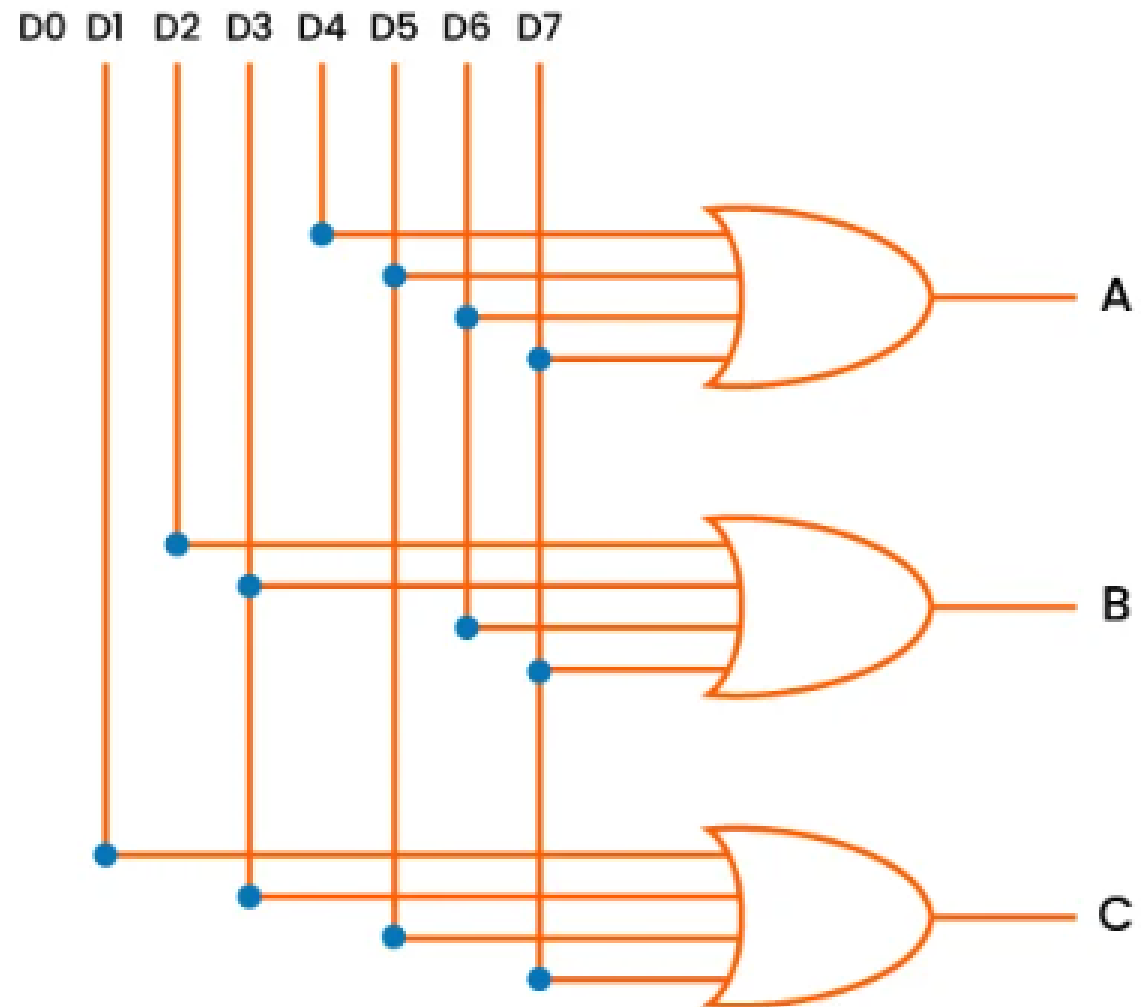
$$C = D1 + D3 + D5 + D7$$

8x3 Encoders

$$A = D4 + D5 + D6 + D7$$

$$B = D2 + D3 + D6 + D7$$

$$C = D1 + D3 + D5 + D7$$



Limitation of Normal Encoders

Let's take the case of this 8-line to 3-line encoder. So, in the case of this 8 to 3 encoder, when this D0 input is high, then the corresponding output is equal to 000. And that is also evident from the logic circuit.

Because here, this D0 input is not connected to any of the OR gates. That means whenever this D0 input is high and all other inputs are zero, then the output of the three OR gates will be equal to 000. But suppose if all the inputs are zero, then also the output of this encoder will be equal to 000. **That means for the two different input conditions, we will get the same output.**

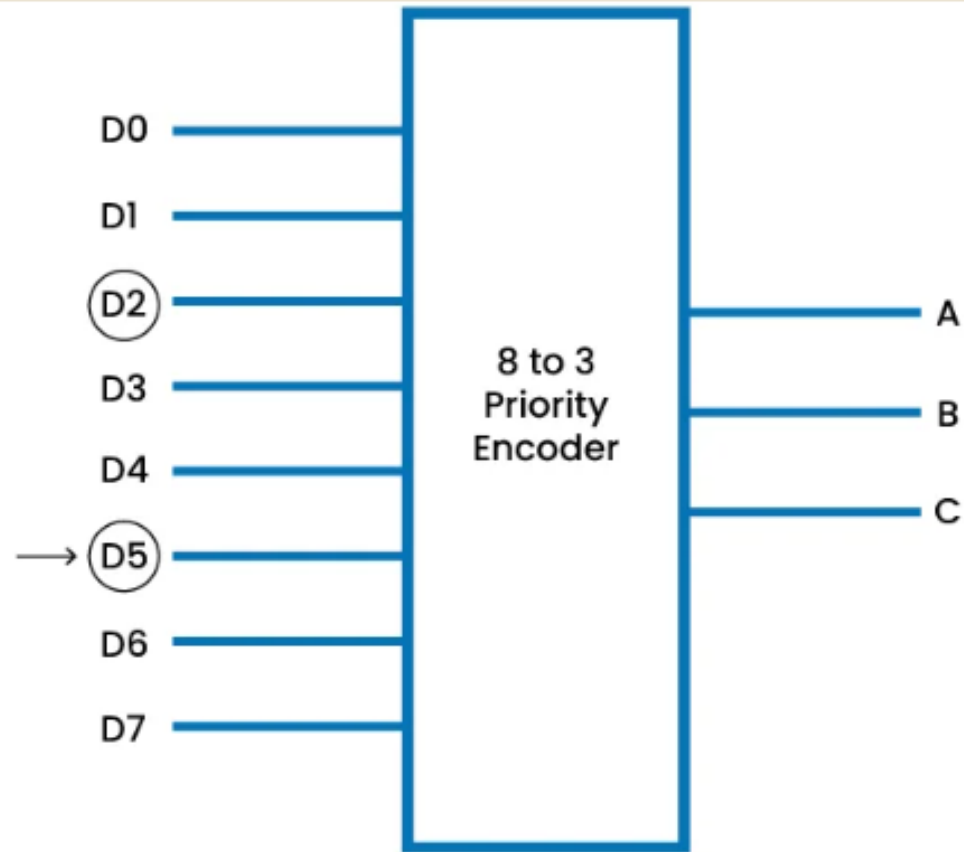
Priority Encoders

“A priority encoder in digital electronics is a combinational circuit that assigns a binary code to the highest-priority active input among multiple inputs. It ensures that when multiple inputs are high, only the one with the highest priority is encoded.”

Priority Encoders

- A priority encoder resolves the problem when multiple inputs are active — it gives priority to the highest-numbered input.

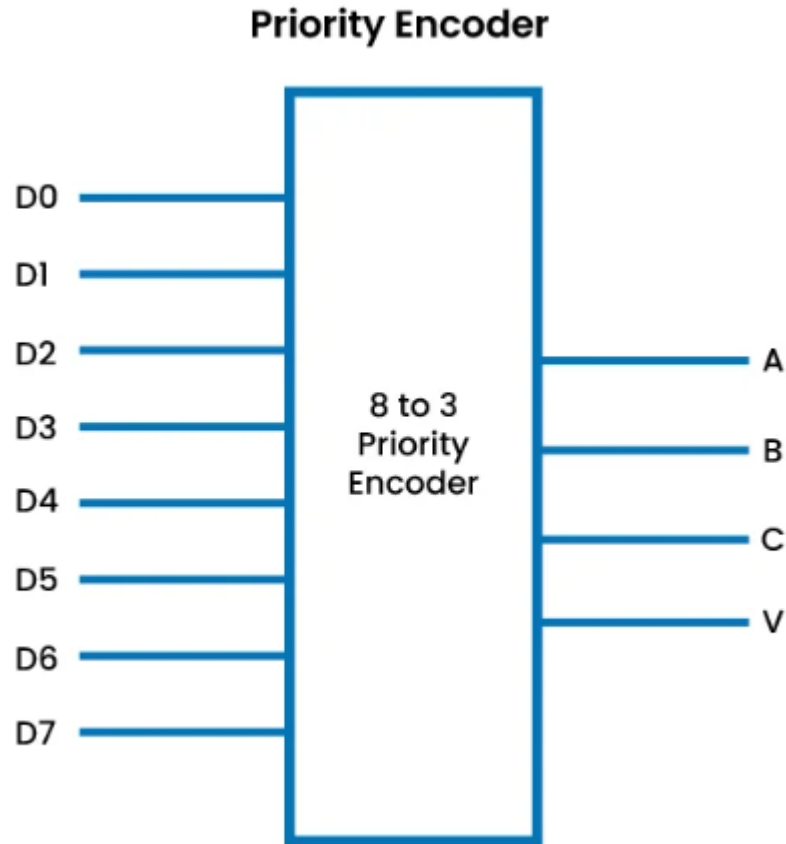
Priority Encoders



For the case, D1 and D2 inputs are high simultaneously; in that case, D2 will have more priority. And in that case, the output of the encoder will be equal to 010.

If D2 and D5 are high simultaneously, in that case, D5 will have more priority. And in that case, the output of the encoder will be equal to 101.

Priority Encoders

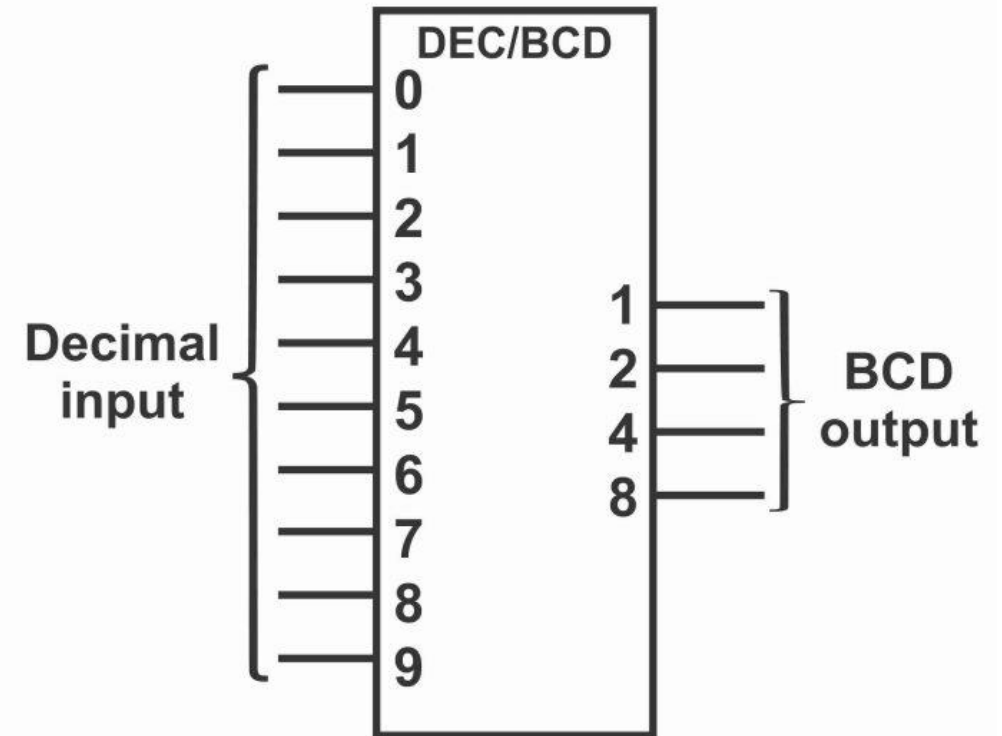


To detect the all-zero input condition, we can also have one more output to this priority encoder. That means when all the inputs are 0, then this output V will also be equal to 0.

While, when one or more than one, input is high, then this output V will be equal to high. So in this way, with the help of the priority encoder, we can eliminate the two limitations of this normal encoder.

Decimal to BCD Encoder

In Decimal to BCD encoder, we have 10 inputs and 4 outputs. So here, each input represents a specific decimal digit, and the output is the BCD equivalent code corresponding to the particular digit.



Decimal to BCD Encoder

Input										Output			
D ₉	D ₈	D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀	Y ₃	Y ₂	Y ₁	Y ₀
0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	0	1	0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0	0	0	0	1	0	1
0	0	0	1	0	0	0	0	0	0	0	1	1	0
0	0	1	0	0	0	0	0	0	0	0	1	1	1
0	1	0	0	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	0	0	0	1	0	0	1

Decimal to BCD Encoder

Truth Table

D0	D1	D2	D3	D5	D6	D6	D7	D8	D9	A	B	C	D
1	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	0	0	0	1	0	0
0	0	0	0	0	1	0	0	0	0	0	1	0	1
0	0	0	0	0	0	1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1	0	0	0	1	1	1
0	0	0	0	0	0	0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	0	0	1	1	0	0	1

$$A = D8 + D9$$

$$B = D4 + D5 + D6 + D7$$

$$C = D2 + D3 + D6 + D7$$

$$D = D1 + D3 + D5 + D7 + D9$$

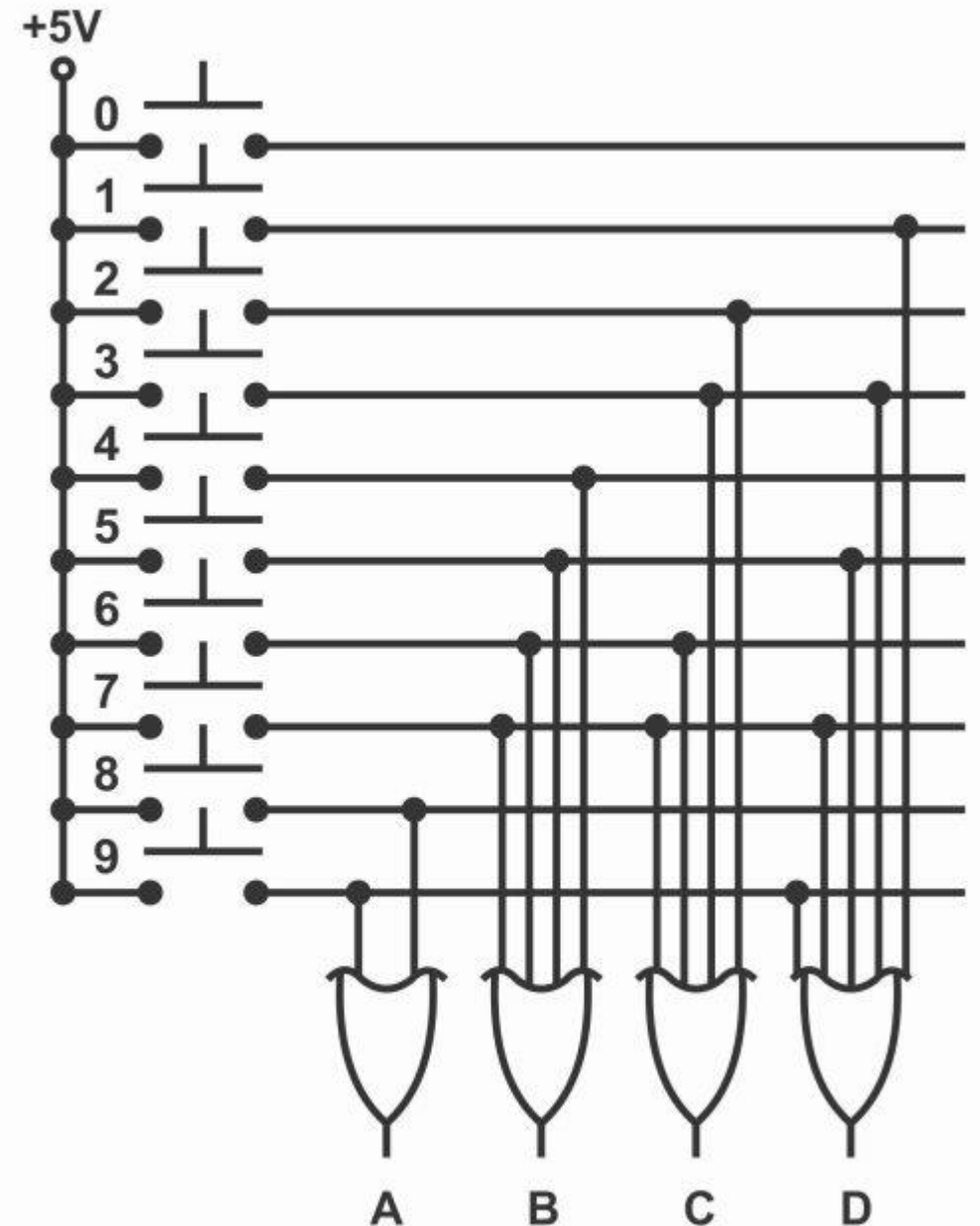
Decimal to BCD Encoder

$$A = D_8 + D_9$$

$$B = D_4 + D_5 + D_6 + D_7$$

$$C = D_2 + D_3 + D_6 + D_7$$

$$D = D_1 + D_3 + D_5 + D_7 + D_9$$



Decimal to BCD Encoder

Zero is not connected to any of the inputs, which causes the problem. Zero is at the output when D0 is active and also when no input lines is selected.

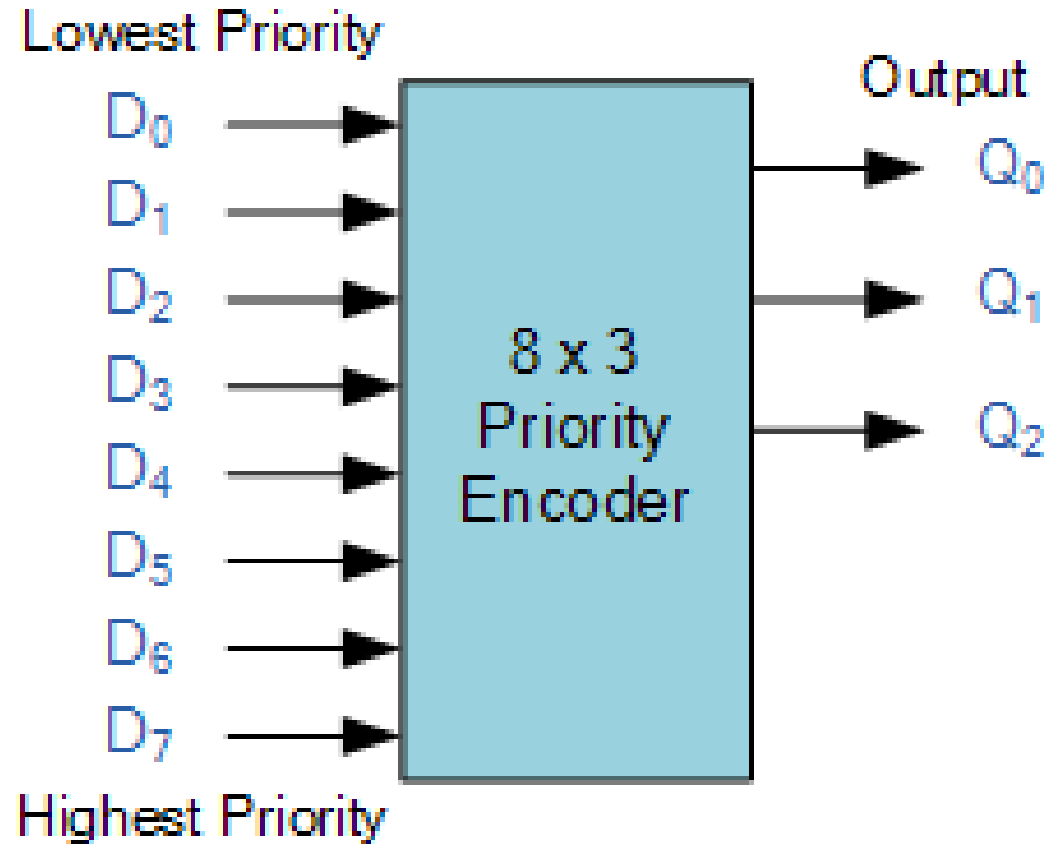
For this a control line 'z' is added to the circuit;
Where z is high, if any of the input is selected.

z is low when none of the input is selected.

This circuit will give errorneouss when more than one line is active simultaneously.

The solution is Priority encoders

8x3 Priority Encoder (IC: 74148)



Inputs								Outputs		
D ₇	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀	Q ₂	Q ₁	Q ₀
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	x	0	0	1
0	0	0	0	0	1	x	x	0	1	0
0	0	0	0	1	x	x	x	0	1	1
0	0	0	1	x	x	x	x	1	0	0
0	0	1	x	x	x	x	x	1	0	1
0	1	x	x	x	x	x	x	1	1	0
1	x	x	x	x	x	x	x	1	1	1

X = dont care

8x3 Priority Encoder

From this truth table, the Boolean expression for the encoder above with data inputs D_0 to D_7 and outputs Q_0 , Q_1 , Q_2 is given as:

$$Q_0 = \Sigma(1, 3, 5, 7)$$

$$Q_0 = \Sigma(\bar{D}_7 \bar{D}_6 \bar{D}_5 \bar{D}_4 \bar{D}_3 \bar{D}_2 D_1 + \bar{D}_7 \bar{D}_6 \bar{D}_5 \bar{D}_4 D_3 + \bar{D}_7 \bar{D}_6 D_5 + D_7)$$

$$Q_0 = \Sigma(\bar{D}_6 \bar{D}_4 \bar{D}_2 D_1 + \bar{D}_6 \bar{D}_4 D_3 + \bar{D}_6 D_5 + D_7)$$

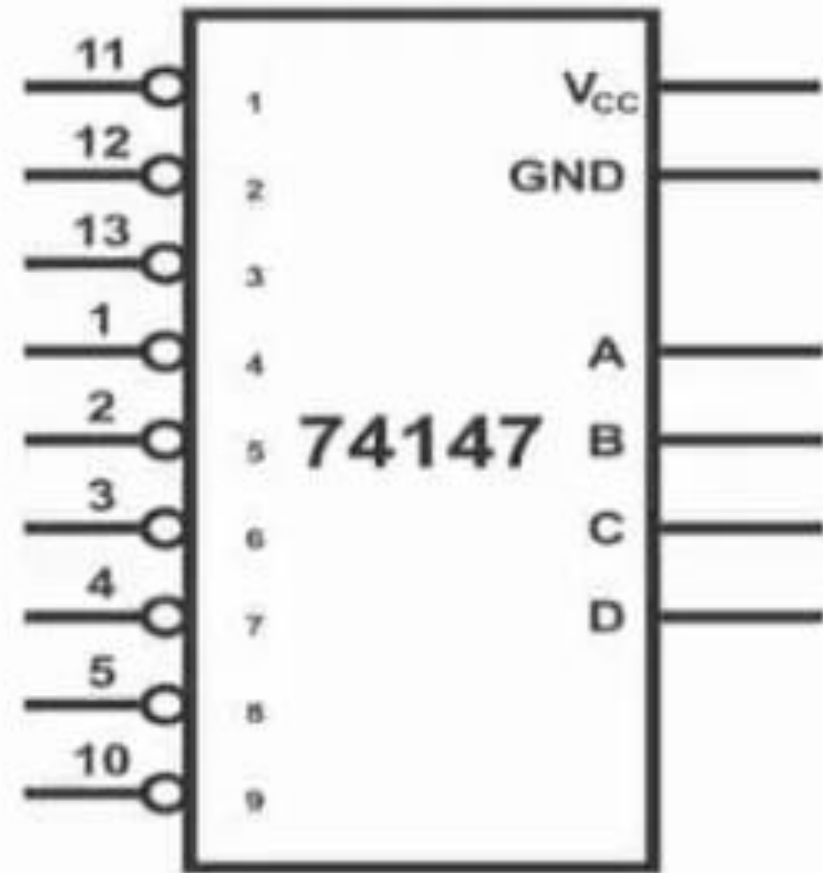
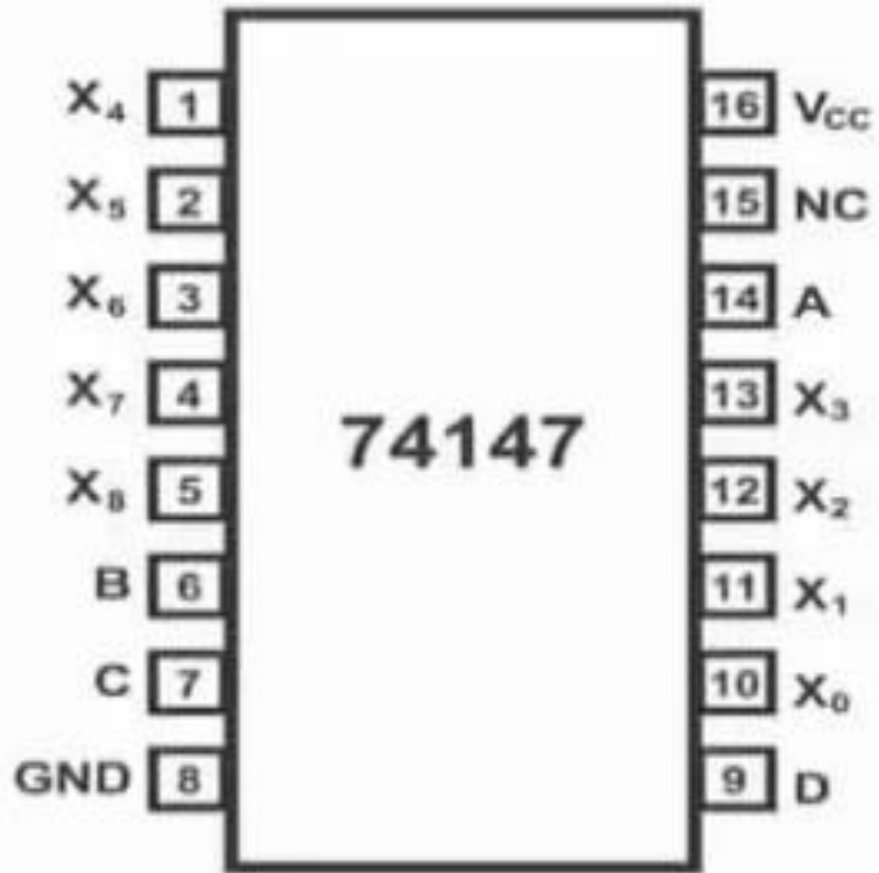
$$Q_0 = \Sigma(\bar{D}_6 (\bar{D}_4 \bar{D}_2 D_1 + \bar{D}_4 D_3 + D_5) + D_7)$$

$$Q_0 = \Sigma(\bar{D}_6 (\bar{D}_4 \bar{D}_2 D_1 + \bar{D}_4 D_3 + D_5) + D_7)$$

$$Q_1 = \Sigma(\bar{D}_5 \bar{D}_4 (D_2 + D_3) + D_6 + D_7)$$

$$Q_2 = \Sigma(D_4 + D_5 + D_6 + D_7)$$

74147 Encoder



Disadvantages of Encoders

- **Increased Complexity:** Encoders can introduce further complexity to digital systems by requiring additional hardware or software. This is applicable for complex or custom encoders.
- **Loss of data:** If the encoders are not constructed right, they may lose data in the process of encoding.
- **Propagation Error:** Misinformation can propagate in the system.
- **Decoding:** The original information must be restored from the encoded data, i.e., decoding or reverse coding is required, which adds an extra step.

Magnitude Comparator

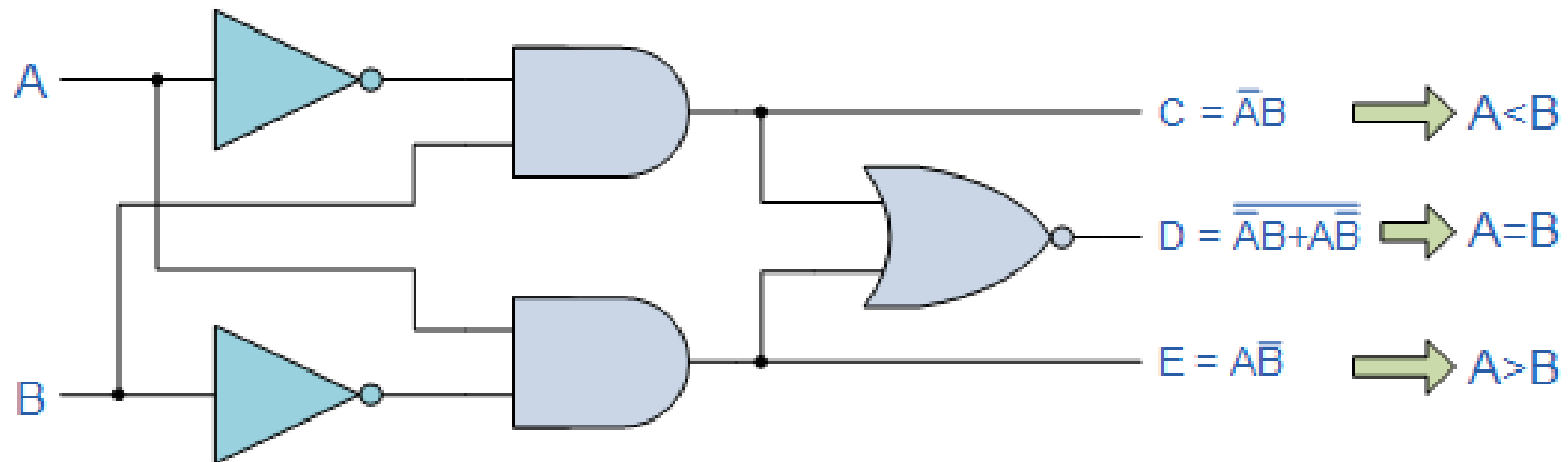
- Magnitude Comparator is a combinational circuit capable of comparing the relative magnitude of two binary numbers.
- It gives output whether one number is greater than the other, or less than or equal. These comparators are used in digital systems, such as for sorting networks, and decision-making circuits to handle numerical comparisons perfectly without any error.



1-bit Magnitude Comparator

A comparator used to compare two bits is called a single-bit comparator. It consists of two inputs each for two single-bit numbers and three outputs to generate less than, equal to, and greater than between two binary numbers. Digital comparators actually use Exclusive-NOR gates within their design for comparing their respective pairs of bits.

The 1-bit comparator is given below.



1-bit Magnitude Comparator

Output $A = B$ is produced when both binary digits are equal in magnitude. That is, either $A = B = "0"$ or $A = B = "1"$.

$A = B$ resembles that of a logic gate function of the Exclusive-NOR, this produces:

$$Q = A \oplus B$$

Inputs		Outputs		
B	A	$A > B$	$A = B$	$A < B$
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

2-bit Magnitude Comparator

It consists of four inputs and three outputs to generate less than, equal to, and greater than between two binary numbers.

Inputs

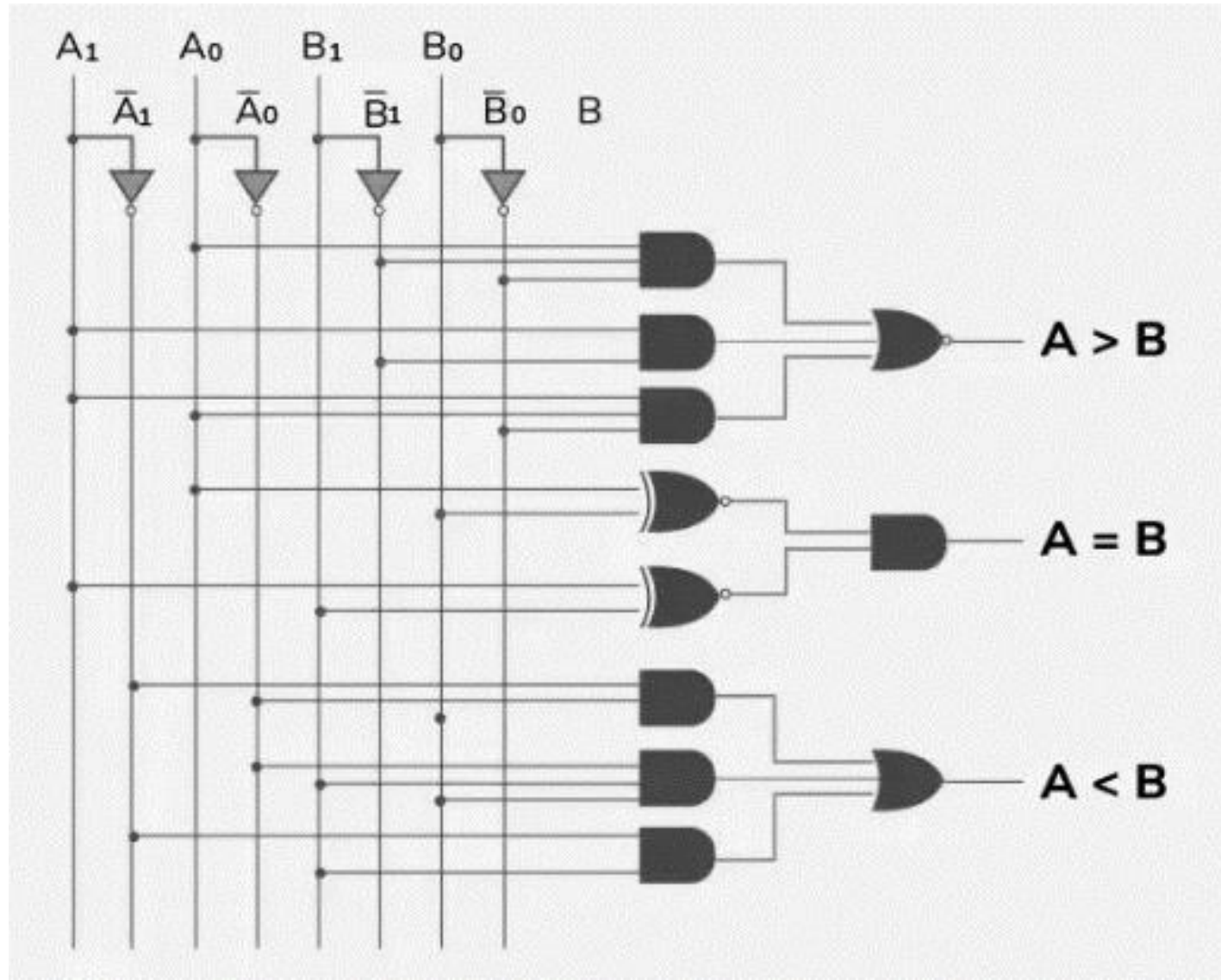
- A1 A0
- B1 B0

Outputs

- $A > B$
- $A = B$
- $A < B$

	A1	A0	B1	B0	A<B	A=B	A>B	
	0	0	0	0	0	1	0	
	0	0	0	1	1	0	0	
	0	0	1	0	1	0	0	
	0	0	1	1	1	0	0	
	0	1	0	0	0	0	1	
	0	1	0	1	0	1	0	
	0	1	1	0	1	0	0	
	0	1	1	1	1	0	0	
	1	0	0	0	0	0	1	
	1	0	0	1	0	0	1	
	1	0	1	0	0	1	0	
	1	0	1	1	1	0	0	
	1	1	0	0	0	0	1	
	1	1	0	1	0	0	1	
	1	1	1	0	0	0	1	
	1	1	1	1	0	1	0	

2-bit Magnitude Comparator



2-bit Magnitude Comparator

Truth Table of Output $A < B$

		$A < B$			
		B_1B_0	00	01	11
A_1A_0	00	0	1	1	1
	01	0	0	1	1
	11	0	0	0	0
	10	0	0	1	0

2-bit Magnitude Comparator

Truth Table of Output A=B

		A = B			
		B1B0	00	01	11
A1A0	00	1	0	0	0
	01	0	1	0	0
	11	0	0	1	0
	10	0	0	0	1

2-bit Magnitude Comparator

Truth Table of Output $A > B$

		A>B				
		B1B0	00	01	11	10
A1A0	00	0	0	0	0	0
	01	1	0	0	0	0
	11	1	1	0	1	0
	10	1	1	0	0	0

2-bit Magnitude Comparator

$$A < B = A_1' B_1' + A_0' B_1 B_0 + A_1' A_0' B_0$$

$$A > B = A_1 B_1' + A_0 B_1' B_0' + A_1 A_0 B_0'$$

A=B:

$$F = A_1' A_0' B_1' B_0' + A_1' A_0 B_1' B_0 + A_1 A_0 B_1 B_0 + A_1 A_0' B_1 B_0'$$

$$F = A_1' B_1' (A_0' B_0' + A_0 B_0) + A_1 B_1 (A_0 B_0 + A_0' B_0')$$

$$F = (A_0 B_0 + A_0' B_0') (A_1 B_1 + A_1' B_1')$$

$$F = (A_0 \text{ Ex-Nor } B_0) (A_1 \text{ Ex-Nor } B_1)$$

$$A = B = (A_0 \text{ Ex-Nor } B_0) (A_1 \text{ Ex-Nor } B_1)$$

4-bit Magnitude Comparator

It consists of eight inputs each for two four-bit numbers and three outputs to generate less than, equal to, and greater than between two binary numbers.

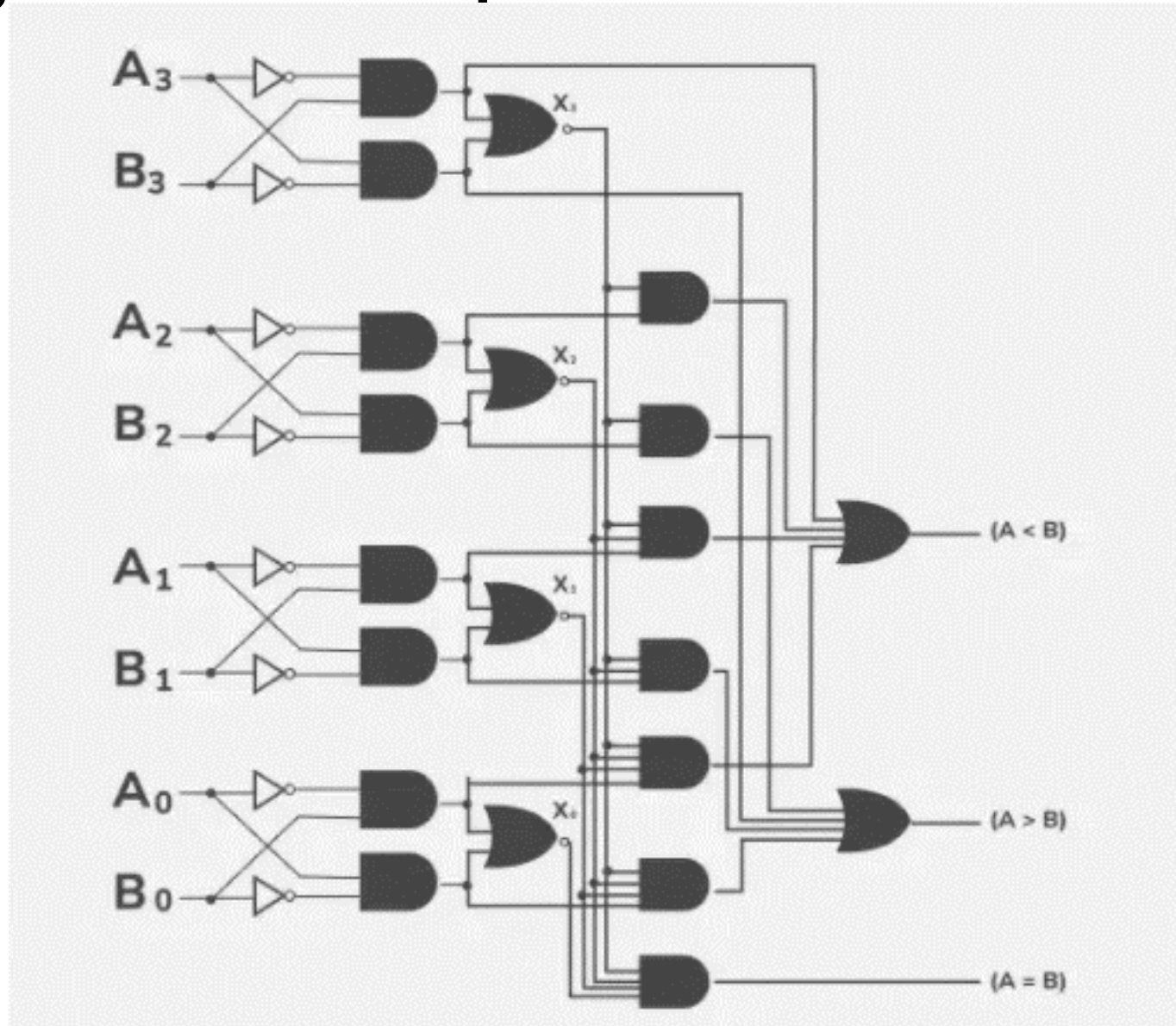
In a 4-bit comparator, the **condition of $A > B$** can be possible in the following four cases.

1. If $A_3 = 1$ and $B_3 = 0$
2. If $A_3 = B_3$ and $A_2 = 1$ and $B_2 = 0$
3. If $A_3 = B_3$, $A_2 = B_2$ and $A_1 = 1$ and $B_1 = 0$
4. If $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$ and $A_0 = 1$ and $B_0 = 0$

Similarly, the **condition for $A < B$** can be possible in the following four cases.

1. If $A_3 = 0$ and $B_3 = 1$
2. If $A_3 = B_3$ and $A_2 = 0$ and $B_2 = 1$
3. If $A_3 = B_3$, $A_2 = B_2$ and $A_1 = 0$ and $B_1 = 1$
4. If $A_3 = B_3$, $A_2 = B_2$, $A_1 = B_1$ and $A_0 = 0$ and $B_0 = 1$

4-bit Magnitude Comparator



Applications of Comparators

- Comparators are used in central processing units (CPUs) and microcontrollers (MCUs).
- These are used in control applications in which the binary numbers representing physical variables such as temperature, position, etc. are compared with a reference value.
- Comparators are also used as process controllers and for Servo motor control.
- Used in password verification and biometric applications.

Advantages & Disadvantages of Comparators

Advantage of Comparator

- Comparators are simple and efficient for comparison of binary values.
- Fast decision-making in Digital Circuits.
- This comparator can be easily integrated in to complex systems like processors and arithmetic units.
- Design of comparator are modular, which allow them to scale solutions for comparing multi-bit numbers

Disadvantages of Comparator

- Comparators have limit of bits for comparison.
- It requires more complex circuit for large bits.
- Power consumption increase with increase in the complexity of the circuit.

Thank you!